

Rain or Ride?

Bike Rental Demand Analysis in Excel

A beginner portfolio project using real public data from Seoul. The objective is to understand when rental demand is high and how demand differs across months, seasons and rainy hours.

Dataset	Project scope
Records	8,760 hourly observations; all retained
Period	1 December 2017 to 30 November 2018
Location	Seoul, South Korea
Analysis population	8,465 operating hours; 295 closed hours excluded
Tools demonstrated	Excel formulas, filters, conditional formatting and four charts

Questions the dashboard answers

1. Which hour has the highest average rentals?
2. Which month has the largest rental volume?
3. How does average hourly demand vary by season?
4. How does average demand differ between rainy and non-rainy hours?

What is included

Rain_or_Ride_Dashboard.xlsx: the working project, with Dashboard and Data tabs.

Seoul_Bike_Dataset.csv: an Excel-friendly version of the complete dataset.

Rain_or_Ride_Project_Guide.pdf: this explanation.

README.md, dashboard preview and original CSV: included in the GitHub ZIP.

Source and reuse

Seoul Bike Sharing Demand [Dataset]. (2020). UCI Machine Learning Repository.
<https://doi.org/10.24432/C5F62R>

Licensed by the source under [Creative Commons Attribution 4.0](#). Keep this credit when uploading to GitHub. The data is historical and is not a description of present-day Seoul. Accessed 19 September 2026.

Understand the workbook

1. Start with the Dashboard tab

The first section shows total rentals, average rentals per operating hour and the busiest hour. Four editable Excel charts show hourly demand, monthly totals, seasonal averages and rain-group averages. The calculation tables begin at row 49.

The Data tab contains the 14 source columns. Filter buttons let you inspect records. These filters do **not** change the dashboard: SUMIFS, COUNTIFS and AVERAGEIFS continue to read the full specified ranges.

2. Follow the calculation flow

Data rows 7-8766 → calculation tables → headline results and charts. All summary calculations remain editable formulas. There are no macros, predictive models or hidden scoring rules.

Headline	How it is calculated
Total rentals: 6,172,314	Sum monthly rental totals in I50:I61. These totals include operating hours only.
Average rentals: 729.2	Total rentals / 8,465 operating hours. A closed hour is not counted in the denominator.
Busiest hour: 18:00	Find the largest hourly average in D50:D73, then return the matching hour label. Ties return the first hour.

3. Read a simple formula

Cell C50 counts operating records where the Hour column equals A50 (0 for midnight):

```
=COUNTIFS('Data'!$C$7:$C$8766,A50,'Data'!$N$7:$N$8766,"Yes")
```

Cell D50 calculates average rentals for the same records:

```
=AVERAGEIFS('Data'!$B$7:$B$8766,'Data'!$C$7:$C$8766,A50,'Data'!$N$7:$N$8766,"Yes")
```

The \$ signs keep the data ranges fixed when a formula is copied. A50 changes to A51 for the next hour. I67 uses the simple division **=I65/I66**.

Monthly formulas include dates on or after the month start and before the next month. EDATE moves the start forward by one month. TEXT creates readable chart labels such as Dec 17.

Results you can explain

Question	Finding
Peak hour	18:00: 1,554.0 rentals per operating hour, averaged across 353 observations.
Busiest month	June 2018: 896,887 rentals across 720 operating hours.
Strongest season	Summer: 1,034.1 rentals per operating hour. Winter: 225.5.
Rain comparison	Rainy hours average 167.3 rentals; non-rainy hours average 765.6.

Why averages matter

There are 7,949 operating hours without rain and only 516 with rain. Comparing group totals alone would be misleading because the group sizes differ. Average rentals per operating hour is a fairer first comparison.

Rain group	Operating hours	Total rentals	Average rentals
No rain (0 mm)	7,949	6,086,009	765.6
Rain (> 0 mm)	516	86,305	167.3

The rainy-hour average is **78.2% lower** than the non-rainy average: $(765.632 - 167.258) / 765.632$. Calculations use unrounded values.

What these findings suggest

Peak-hour preparation: check bike availability before the evening peak. The dataset does not contain station-level inventory, so it cannot tell us how many bikes to move or where to move them.

Seasonal planning: plan extra availability checks during higher-demand periods. Monthly totals describe overall activity; month length and operating hours also affect those totals.

Weather awareness: include weather when planning operations, but do not treat the 78.2% difference as a causal estimate. Rainy and non-rainy hours may also differ in temperature, season and time of day.

Claims to avoid

Do not say these are unique customers, that all evening riders are commuters, or that rain causes an exact 78.2% reduction. Do not generalize one historical year in Seoul to all cities.

Dataset and preparation

One row describes one hour for the Seoul bike-sharing system. Rented Bike Count counts rentals during that hour; the same person or bicycle may appear in multiple rentals.

Column	Meaning / unit
Date	Calendar date; supplied as DD/MM/YYYY
Rented Bike Count	Number of rentals during the hour
Hour	Hour of day, 0-23
Temperature (C)	Air temperature in degrees Celsius
Humidity (%)	Humidity reported on a 0-100 scale
Wind speed (m/s)	Wind speed in metres per second
Visibility (10m)	Value represents units of 10 metres
Dew point (C)	Dew point temperature in Celsius
Solar radiation (MJ/m2)	Solar radiation measurement
Rainfall (mm)	Rainfall in millimetres
Snowfall (cm)	Snowfall in centimetres
Seasons	Winter, Spring, Summer or Autumn
Holiday	Holiday or No Holiday
Functioning Day	Yes = operating; No = not operating

Preparation performed

1. Read the original CSV using its Windows-1252 encoding.
2. Parse dates as day/month/year; store real dates in Excel and ISO dates in the cleaned CSV.
3. Keep numbers numeric and simplify column labels.
4. Check blanks and repeated Date + Hour combinations: both checks returned zero.
5. Keep all records and use Functioning Day = Yes in demand summaries.

No missing values were filled, no unusual observations were deleted, and no data was invented. All 295 non-operating records have zero rentals. The ZIP preserves the downloaded original CSV unchanged.

Use it for your project

A simple way to present it

“My project analyses bike-rental demand using 8,760 hourly records from Seoul. I used Excel to compare demand by hour, month, season and rainfall. I excluded closed hours from demand averages and kept the original records. The busiest hour was 18:00, and summer had the highest seasonal average. Rainy hours had lower average rentals, but the comparison does not prove cause and effect.”

What to practise before submitting

1. Open Dashboard and explain the three headline numbers.
2. Scroll to row 49 and click C50 and D50 to read their formulas.
3. Explain why an average is more useful than a total for rain groups.
4. Open Data, filter Functioning Day to No, then clear the filter.
5. Be able to name one limitation of the analysis.

Put the project on GitHub

Create a repository named **rain-or-ride-excel-analysis**. Extract the supplied ZIP and upload its files, keeping README.md at the repository root. The README includes the objective, results, skills, preview and dataset credit. GitHub can show the README, CSV and image; visitors can download the XLSX to use the dashboard.

Suggested repository description: **Beginner Excel analysis of Seoul bike-rental demand by hour, season and rainfall using a public UCI dataset.**

Editing and updating

Changes within Data rows 7-8766 recalculate the summary formulas and charts. Excel calculation mode should be Automatic. The workbook is designed for this fixed historical dataset. If you append records, extend every summary range beyond row 8766 and add any new months to the monthly table and chart. Data filters are for inspection only.

Verification and limitations

Headline results and hourly averages were checked against independent calculations from the CSV. Formula results were scanned for errors, and a temporary source edit was used to check recalculation before restoring the original value. Charts and document pages were visually reviewed. The file was not opened in desktop Microsoft Excel during creation.

The project is descriptive: it does not forecast future demand, estimate revenue, identify customers or specify station-level bike requirements. Learn the formulas before presenting the work as your own analysis.